



ACHARYA INSTITUTE OF TECHNOLOGY

Affiliated to Visvesvaraya Technological University, Belagavi, Govt. of Karnataka.

Approved by AICTE, New Delhi

Computer Science & Engineering (Data Science)

PROJECT REPORT ON

From BI to Big Data: Explain, Design & Defend

Subject Name: Big Data Analytics

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1. About the Project

1.1 Introduction to Big Data

In today's digital world, data is constantly created through activities like video streaming, social media, online shopping, and digital payments. Each click, search, view, or interaction generates information. When this information grows very large, is produced at great speed, and comes in various formats, it is referred to as Big Data.

Big Data has five key characteristics, known as the 5 V's:

- **Volume:** This refers to the huge amounts of data created daily, measured in terabytes and petabytes.
- **Velocity:** This indicates how fast data is generated and processed in real time.
- **Variety:** Data comes in different forms, including structured data (like tables), semi-structured data (like logs), and unstructured data (like videos, images, and text).
- **Veracity:** This represents how reliable and accurate the data is.
- **Value:** This is about the meaningful insights derived from data that help in making decisions.

Traditional data processing systems cannot efficiently handle such large, complex, and rapidly changing data. Thus, modern Big Data technologies are necessary to store, process, and analyze this information effectively.

1.2 Overview of Netflix as a Big Data Application

Netflix is a leading video streaming platform that offers movies, television series, and original content to millions around the world. It operates in over 190 countries and has a vast number of subscribers.

Every second, Netflix produces vast amounts of data, which includes:

- User watch history
- Search queries
- Clickstream data
- Pause, rewind, and skip actions
- Ratings and reviews
- Device information
- Subscription and billing details

This data is continuously created and stored. The amount of information is enormous because millions of users access the platform at the same time. Handling such large and varied data requires distributed storage systems and fast processing engines.

1.3 Why Netflix Requires Big Data

Netflix relies on Big Data analytics to enhance user experience and improve business performance. Key reasons include:

1. Personalized Recommendation System

Netflix examines user behavior and viewing habits to suggest content that fits individual tastes. This personalization boosts user engagement and satisfaction.

2. Content Strategy and Production Decisions

By analyzing viewing trends and audience preferences, Netflix discovers what types of content are popular. Big Data informs the decision on which new shows or movies to create.

3. Real-Time Streaming Optimization

Netflix employs real-time analytics to adjust video quality based on network speed and device capability, ensuring users enjoy smooth streaming.

4. Customer Retention and Churn Prediction

Using predictive analytics, Netflix identifies users who might cancel their subscriptions and can take steps such as offering personalized suggestions or promotional deals.

Without Big Data technologies, processing millions of user interactions and generating insights in real time would be very challenging.

1.4 Limitations of Traditional Business Intelligence

Traditional Business Intelligence (BI) systems mainly handle structured data stored in relational databases. While they work well for small and medium-sized businesses, they have several shortcomings when used by large streaming platforms like Netflix.

Some major limitations include:

- Inability to process unstructured data like video logs and user interaction logs.
- Limited scalability for handling petabytes of data.
- Slow processing speed for real-time analytics.
- High costs associated with scaling traditional database systems.

Given these challenges, traditional tools like Excel and basic SQL databases cannot meet Netflix's massive data needs. Therefore, modern Big Data frameworks and distributed computing systems are essential.

2. Tools and Technologies Used

2.1 Apache Hadoop

Apache Hadoop is an open-source framework used to store and process large amounts of data across multiple machines. It is designed for distributed computing and handles Big Data efficiently.

Hadoop mainly consists of:

- HDFS (Hadoop Distributed File System) – Stores large datasets by dividing them into blocks and distributing them across multiple nodes. This ensures scalability and fault tolerance.
- MapReduce – A programming model used for processing large datasets in parallel. It divides tasks into smaller parts and processes them simultaneously.

In Netflix, Hadoop can store massive user logs, viewing history, and streaming data.

2.2 Apache Spark

Apache Spark is a fast data processing engine that performs in-memory computation, making it much faster than traditional MapReduce.

Key features:

- In-memory processing
- Real-time analytics
- Batch and streaming support
- Machine learning support through MLlib

For Netflix, Spark is used to analyse user behaviour, detect trending shows, and power the recommendation system.

2.3 NoSQL Database (Cassandra)

Apache Cassandra is a distributed NoSQL database designed to handle large-scale data across multiple servers.

Unlike traditional relational databases, Cassandra:

- Does not require fixed schemas
- Supports high scalability
- Provides high availability

Netflix uses NoSQL databases to store user profiles, watch history, and session data.

2.4 Business Intelligence Tools

Business Intelligence tools help visualize and analyse processed data.

Examples:

- Microsoft Power BI
- Tableau

These tools create dashboards showing:

- Daily active users
- Most watched content
- Revenue trends
- Regional performance

They help management make data-driven decisions.

2.5 Machine Learning

Machine Learning is used to build Netflix's recommendation system.

It helps in:

- Personalized content suggestions
- Predicting user churn
- Identifying trending genres
- Improving user engagement

Spark MLlib and other AI frameworks are used to develop predictive models.

3. Architecture Design

3.1 Traditional Data Warehouse Architecture

In traditional systems, data is stored in relational databases and processed using structured query methods.

Architecture Flow

Traditional Data Warehouse Architecture - Netflix Case Study

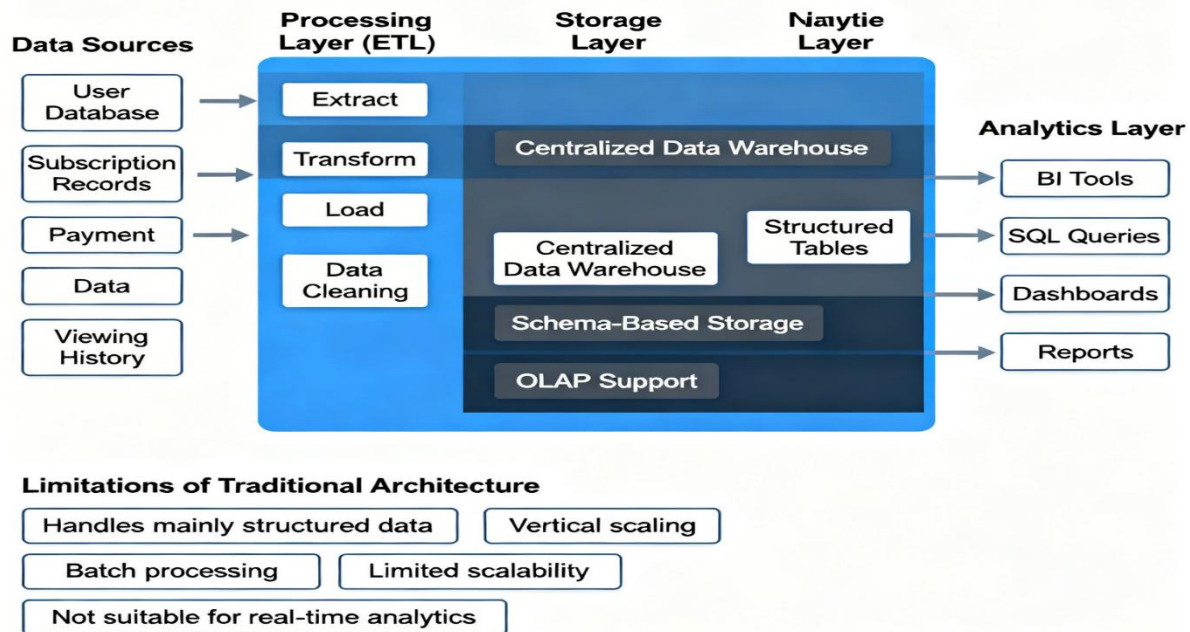


Fig 3.1 – Architecture Design

1. Data Sources

For Netflix, data sources include:

- User watch history
- Subscription data

- Payment information
- Ratings and reviews

2. ETL (Extract, Transform, Load)

- Extract data from sources
- Transform into structured format
- Load into Data Warehouse
- This process is usually batch-based (not real-time).

3. Data Warehouse

Centralized storage system (SQL-based).

Stores structured data in tables.

Limitations:

- Cannot store unstructured video logs efficiently
- Scaling is expensive
- Slow for large datasets

4. BI Tools

Tools like:

- Microsoft Power BI
- Tableau

Used to generate reports like:

- Monthly subscriptions
- Revenue reports
- Most watched movies

3.2 Hadoop-Based Big Data Architecture

Architecture Flow

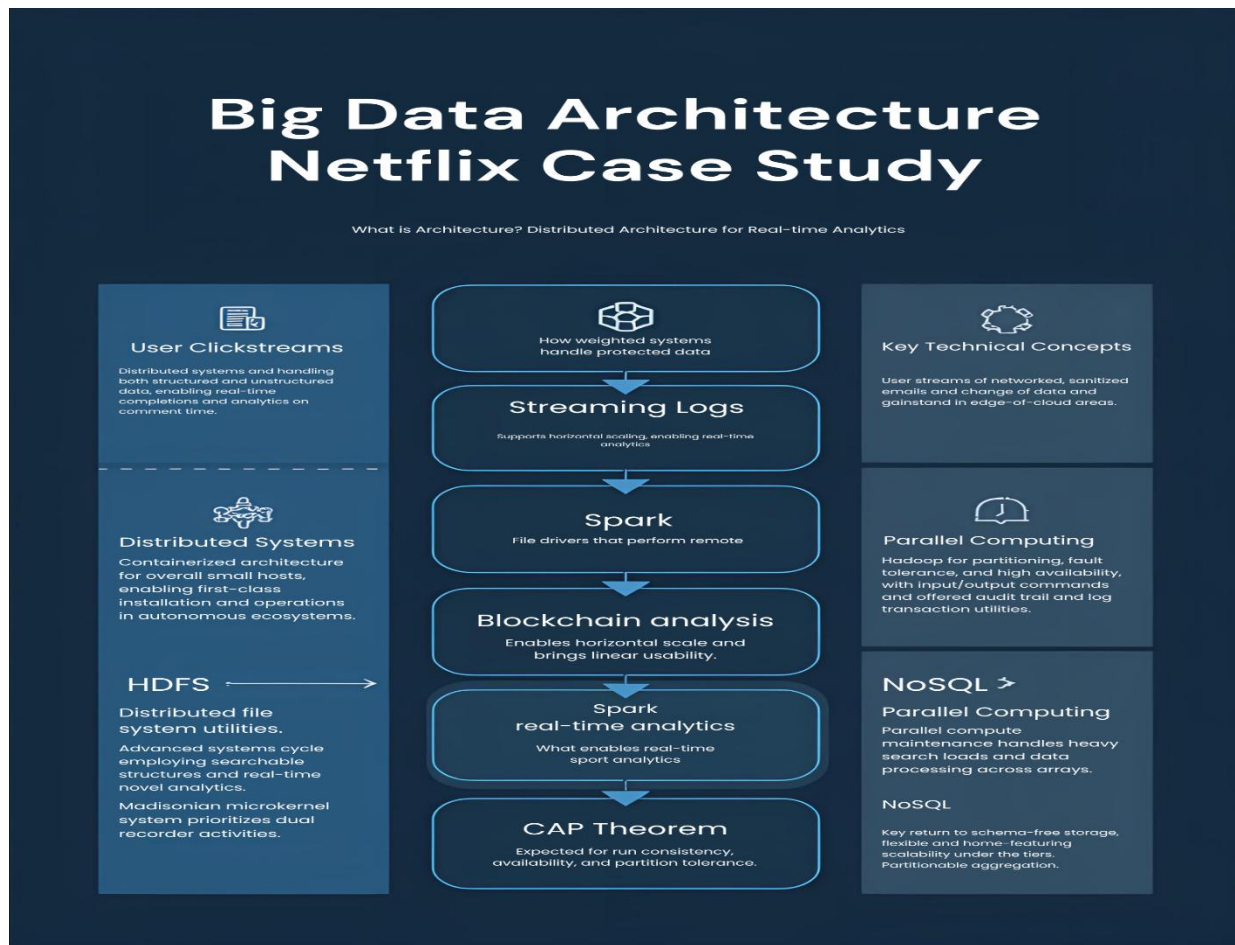


Fig 3.2 – Hadoop Architecture

1.Data Sources

- User clicks
- Watch history
- Streaming logs

- Device data
- Search queries

Massive data generated every second.

2.Data Ingestion: Data is collected using streaming systems and loaded into distributed storage.

3.HDFS (Storage Layer)

Apache Hadoop

Stores massive datasets across multiple nodes.

Advantages:

- Distributed storage
- Fault tolerance
- Scalability

4. Processing Layer

Apache Spark

Processes data in-memory.

Used for:

- Real-time analytics
- Trend detection
- Behaviour analysis

5. NoSQL Database

Apache Cassandra

Stores processed user profiles and recommendation data.

4. BI vs Big Data – Role Play

Conversation

Manager: Why do we need Big Data? We already use Excel and SQL. That should be enough.

Consultant: Excel and SQL work well for small datasets. But Netflix handles data from millions of users every second. Traditional tools cannot manage that scale.

Manager: But we already generate monthly reports using BI tools. What more do we need?

Consultant: Monthly reports only tell us what happened. Netflix needs real-time insights, like what users are watching right now and what content is trending globally.

Manager: Can't we just upgrade our database server?

Consultant: Even a powerful single server has limitations. Netflix generates petabytes of structured and unstructured data such as video logs, clickstreams, and viewing patterns. A single machine cannot handle that efficiently.

Manager: What exactly is the problem with traditional BI systems?

Consultant: Traditional BI systems are designed for structured data stored in relational tables. Netflix data includes unstructured data like streaming logs and user behaviour data, which cannot be handled efficiently in traditional databases.

Manager: So, what is the alternative?

Consultant: Big Data technologies like Hadoop and Spark uses distributed computing. Instead of storing data in one system, they distribute it across multiple machines.

Manager: How does that help?

Consultant: It increases scalability and speed. If data increases, we simply add more machines. Processing also becomes faster because tasks run in parallel.

Manager: What about user recommendations? How are those generated?

Consultant: That is done using Machine Learning models built on Big Data frameworks. These models analyse user watch history and predict what content the user will like next.

Manager: Can traditional SQL not do predictions?

Consultant: SQL can perform basic analysis, but predictive analytics and real-time recommendation systems require advanced processing engines like Spark MLlib.

Manager: What business benefits does this provide?

Consultant: It improves user experience, increases customer retention, reduces churn, and helps Netflix decide which shows to produce based on data insights.

Manager: So Big Data is not just about storage?

Consultant: Exactly. It is about intelligent decision-making, scalability, and real-time analytics.

Manager: I understand now. Excel and traditional BI are not sufficient for a platform like Netflix.

Consultant: Yes. For small businesses, BI is enough. But for global platforms like Netflix, Big Data is essential.

5. Analytics & Tool Match

In Big Data environments like Netflix, different types of analytics are used to answer different business questions. Each analytics type uses specific tools and technologies.

Descriptive Analytics

Answers: What happened?

- Summarizes historical data using reports and dashboards.
- Identifies trends and patterns from past data.

Diagnostic Analytics

Answers: Why did it happen?

- Analyses data to find the root cause of a problem or trend.
- Uses comparison and correlation analysis to explain changes.

Predictive Analytics

Answers: What will happen next?

- Uses historical data and machine learning to forecast future outcomes.
- Identifies patterns to predict trends and user behaviour.

Prescriptive Analytics

Answers: What action should be taken?

- Recommends the best action based on predictive results.
- Uses optimization and AI models to improve decision-making.

Business Question	Analytics Type	Tool Used
What happened?	Descriptive Analytics	Hadoop + BI Tools
Why did it happen?	Diagnostic Analytics	Spark + Data Mining
What will happen next?	Predictive Analytics	Spark MLlib / Machine Learning
What action should be taken?	Prescriptive Analytics	AI Recommendation Engine + NoSQL

Table 5.1 – Analytics Tool

Analytics helps convert large amounts of Netflix data into useful insights. Descriptive, diagnostic, predictive, and prescriptive analytics work together to understand past trends, identify causes, predict future outcomes, and suggest actions. Using Big Data tools, Netflix improves user experience and makes better business decisions.

6. Contribution & Implementation

6.1 Project Contribution

This project was carried out as a theoretical study to understand the application of Big Data technologies in real-world streaming platforms such as Netflix. The primary contribution of this work includes research, analysis, architecture design, and comparative study of traditional Business Intelligence systems and modern Big Data frameworks.

The major contributions in this project are as follows:

- Studied the concept of Big Data and its characteristics (5 V's).
- Analysed how Netflix generates and manages massive volumes of data.
- Compared Traditional BI systems with Big Data architectures.
- Designed two architectures:
 - Traditional Data Warehouse Architecture
 - Hadoop-Based Big Data Architecture
 - Classified analytics into descriptive, diagnostic, predictive, and prescriptive types.
 - Developed a role-play explanation to clearly differentiate BI and Big Data.

This project demonstrates a clear understanding of how large-scale data systems operate in modern digital platforms.

6.2 Implementation Approach

Although this project does not include practical coding implementation, it focuses on conceptual and architectural design based on real-world Big Data practices.

The implementation approach followed in this project includes:

1. Identifying Netflix as a real-world Big Data application.
2. Studying the data flow from user interaction to analytics generation.
3. Designing a traditional architecture to understand its limitations.
4. Designing a Hadoop-based Big Data architecture to show scalability and real-time processing.
5. Mapping analytics types to appropriate Big Data tools.

The architectures were designed to clearly represent data sources, storage layers, processing layers, machine learning layers, and visualization components.

This structured approach ensures a clear understanding of how Big Data technologies are integrated into a streaming platform like Netflix.

7. Results

The overall result of this project confirms that Big Data architecture is significantly more efficient, scalable, and intelligent compared to traditional BI systems. For global streaming platforms like Netflix, Big Data is not optional but essential for sustainable growth and innovation.

The primary outcome of this project is the comparative analysis between Traditional Data Warehouse Architecture and Hadoop-Based Big Data Architecture in the context of Netflix. The project also classified analytics into four categories: descriptive, diagnostic, predictive, and prescriptive analytics.

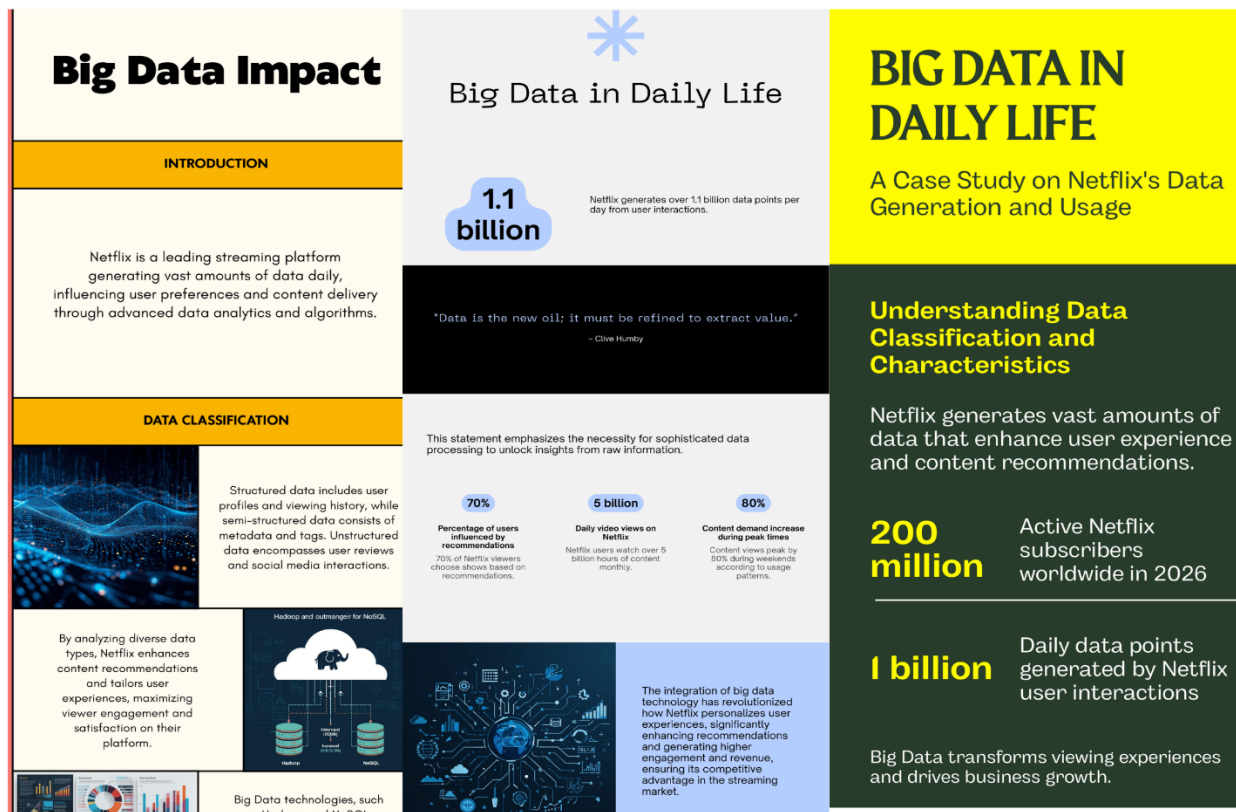


Fig 7.1 – Infographic poster (Big data in Daily Life)



Fig 7.2 – Traditional Data Warehouse

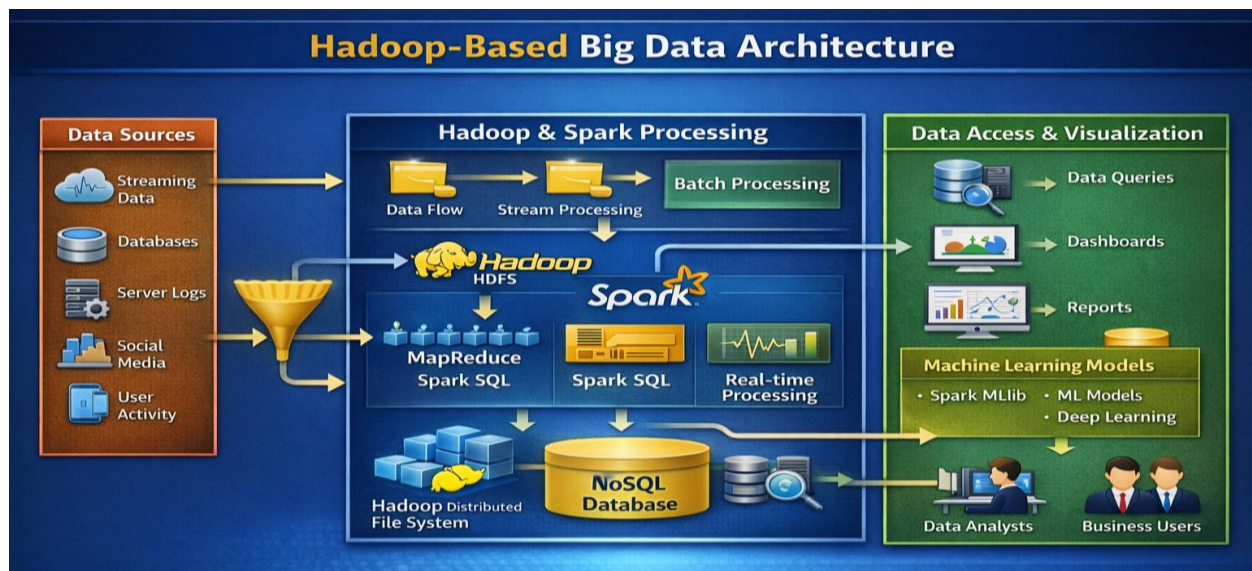


Fig 7.3 – Hadoop based Big Data

8. Conclusion

This project provided a comprehensive understanding of how Big Data technologies are applied in real-world streaming platforms such as Netflix. Through detailed analysis, the study highlighted the limitations of traditional Business Intelligence systems and emphasized the need for scalable and distributed Big Data frameworks. It was observed that conventional data warehouse architectures are suitable only for structured and small-scale data, whereas modern applications generate massive volumes of structured and unstructured data in real time.

The comparison between Traditional Architecture and Hadoop-Based Big Data Architecture clearly demonstrated the advantages of distributed storage and parallel processing systems. Technologies such as Hadoop and Spark enable efficient data handling, faster computation, and real-time analytics, which are essential for platforms serving millions of users simultaneously.

Furthermore, the classification of analytics into descriptive, diagnostic, predictive, and prescriptive categories helped in understanding how data is transformed into actionable insights. Netflix utilizes these analytics techniques to improve user engagement, personalize recommendations, optimize streaming quality, and make strategic business decisions.